1.Project title: Big Data Analysis of Healthcare Data Using Apache Spark and MongoDB

Team Members:

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2.Dataset choice

We are using a healthcare dataset provided in CSV format obtained from Kaggle (https://www.kaggle.com/datasets/priyamchoksi/100000-diabetes-clinical-dataset).

We chose this dataset because healthcare data is very important in real life and is often large and complex. It has important health data such as age, BML, glucose level, hypertension, and diabetes. It is suitable for Big Data analysis and helps us understand how data can be used to improve decision-making in healthcare systems especially in diabetes.

3. Problem Statement

Healthcare organizations generate a large amount of data every day, but it is not easy to analyse this data using traditional methods. The data can be complex and time-consuming to process.

The problem we are addressing is how to efficiently store, process, and analyse healthcare data using Big Data tools. This is important because better data analysis can help improve healthcare services and decision-making. It helps to identify key risk factors for diabetes and how lifestyle and medical conditions affect diabetes.

4. Project Objectives

The main objectives of this project are:

- To store the healthcare dataset in MongoDB using a proper structure
- To process and analyse the data using Apache Spark (PySpark)
- To perform queries using Spark SQL and MongoDB
- To compare the performance of both tools
- To improve query performance using indexing in MongoDB
- To visualize the results using charts and graphs

5. Expected Outcomes

From this project, we expect to find useful insights from the healthcare data, such as patterns in patient information and healthcare services. We also expect to see insights how age, lifestyle, medical conditions contribute to diabetes.

We also expect to understand how Spark and MongoDB work in handling large datasets and compare their performance. The results will be presented using simple visualizations to

make them easy to understand. We also expect to identify patterns and trend in healthcare data.